

Avish Jhalani

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EXPERIENCE

Global AI Technologies | Role: Software Engineering Intern

Jan 2026 - June 2026

Tech Stack: Python (NumPy, Pandas, Matplotlib), MySQL, Excel, Power BI

- Conducted end-to-end analysis of **100,000+ row** Q4 e-commerce transaction dataset using Python (Pandas, NumPy, Matplotlib) and MySQL, applying hypothesis testing (**Chi-square tests for categorical association, correlation analysis**) to identify the root causes behind a return rate spike from **12% to 19%**.
- Utilized **cohort and segment-wise statistical comparison** to isolate the two highest-risk drivers: COD orders (**84.4% return rate vs. prepaid orders**, validated via **Chi-square test of independence**) and size-mismatch returns (**37% of total**, confirmed as a significant contributor via proportion z-test).
- Designed and built an interactive **Power BI dashboard** from raw CSV data — cleaning, modelling relationships, and visualizing key return trends translating insights into operational recommendations.

PROJECTS

PowerPilot AI | Python, FastAPI, DuckDB, Polars, Groq LLM, Docker

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- Architected an autonomous, agentic pipeline that ingests raw CSV/Parquet, cleans, models, and compiles Power BI dashboards end-to-end, processing **500,000+ records in 8.6s at <35MB peak RAM** via zero-copy **DuckDB profiling and out-of-core Polars streaming**.
- Built an LLM-driven semantic modeling agent (Groq gpt-oss-120b) that autonomously synthesizes **6-8 production-grade DAX** measures from natural-language business goals within a strict **4,000-character context budget**.
- Designed a sandboxed, **self-healing execution layer** for AI-generated code using AST-based static analysis to block unsafe system calls, with a **3-attempt autonomous error-recovery loop**, achieving zero unsafe executions across all production runs.

Collab-Docs | React, NestJS, Yjs, Socket.io, PostgreSQL, Redis, Prisma, Docker

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- Engineered a real-time collaborative document workspace using **Yjs CRDTs** and **WebSockets**, deployed across Vercel, Render, and Supabase, achieving **327ms** average round-trip sync latency on cross-continent connections.
- Reduced PostgreSQL write overhead by **98%** by replacing per-keystroke writes with a custom **5-second debounced** state-save buffer.
- Cut network payload size by **90%+** by transmitting edits as compressed binary **Yjs deltas (30-60 bytes/frame)** instead of full document state, using a **Redis Pub/Sub** adapter to sync state across load-balanced instances.

BloodLink | Next.js, NestJS, PostgreSQL, Redis, Prisma, JWT, Nodemailer

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- Built a full-stack donor-matching platform (Next.js frontend, NestJS/Prisma backend) that geolocates and matches blood requests to eligible donors within a 10km radius using **PostGIS spatial queries** (ST_DWithin, ST_Distance).
- Accomplished verified reliability of the core donor-matching flow, as measured by an **11-case E2E suite** passing against a seeded PostgreSQL database, by writing Jest/Supertest integration tests covering auth, matching, confirmation, and fulfilment.
- Secured sessions using JWT stored in httpOnly, sameSite cookies, protecting against **XSS-based token theft**.

TECHNICAL SKILLS

Programming Languages: C++, JavaScript, Python, SQL

Web & Cloud Technologies: Node.js, Express.js, Next.js, RESTful API, Socket.io, FastAPI

Database Systems & Tools: PostgreSQL, Prisma, Redis, Git/GitHub, Postman, Docker

EDUCATION

JECRC University, Jaipur

2022-2026

Bachelor of Technology - Computer Science

CGPA: 8.7

- Coursework: Object-Oriented Programming, Data Structures, Algorithms, Operating Systems, Software Engineering, Database Management Systems, Computer Networks, System Design

ACHIEVEMENTS

- Solved over 500 problems across platforms, ranking in the top 14% in LeetCode Biweekly Contest 187